



Lesson Plan

Pointers

Q. What is a pointer ?

The pointer is a variable, it is also known as locator or indicator that points to an address of a value. The symbol of an address is represented by a pointer. In addition to creating and modifying dynamic data structures, they allow programs to emulate call-by-reference.

For example, *pointer_name

Dereferencing operator(*) : Dereferencing is the method where we are using a pointer to access the element whose address is being stored. We use the * operator to get the value of the variable from its address.

```
int a;
// Referencing
int *ptr=&a;
// Dereferencing
int b=*ptr;
```

Note : Arithmetic such as add, subtract can be performed on pointers as well.

Pointer Increment in C++:

```
int array[5]={2,4,6,8,11}
```

```
int *ptr = array;
```

```
ptr++; //pointer increment won't add 1 to address instead it move to the next immediate index.
```

Pointer Decrement in C++ :

```
int array[5]={1,2,3,4,5}
```

```
int *ptr=array;
```

```
ptr--;
```

```
ptr--; //pointer decrement will decrement to one less index in the array therefore we have first incremented to the next index so that it might not result in runtime error.
```

Similarly, pointers can be added or subtracted by any constant value as well.

Double pointers : A pointer stores the memory address of other variables. So, when we define a pointer to a pointer, the first pointer is used to store the address of the variables, and the second pointer stores the address of the first pointer. For this very reason, this is known as a Double Pointer or Pointer to Pointer.

For example,

```
#include <bits/stdc++.h>
using namespace std;
int main(){
int data = 1;
int* pointer1;
int** pointer2;
pointer1 = &data;
pointer2 = &pointer1;
cout << "Value of data :- " <<data << endl;
cout << "Value of data using single pointer :- " <<
```

```
*pointer1 << endl;
cout << "Value of data using double pointer :- " <<
**pointer2 << endl;
return 0;
}
```

Output :

Value of data :- 1
 Value of data using single pointer :- 1
 Value of data using double pointer :- 1

Q1. Write a program to calculate sum of two numbers using pointers.

```
#include <iostream>

// Function to calculate the sum of two numbers using pointers
void calculateSum(int* num1, int* num2, int* result) {
    *result = *num1 + *num2;
}

int main() {
    int num1, num2, sum;

    // Input two numbers
    std::cout << "Enter the first number: ";
    std::cin >> num1;

    std::cout << "Enter the second number: ";
    std::cin >> num2;

    // Pointers to the numbers and result
    int* ptrNum1 = &num1;
    int* ptrNum2 = &num2;
    int* ptrSum = &sum;

    // Call the calculateSum function using pointers
    calculateSum(ptrNum1, ptrNum2, ptrSum);

    // Print the sum
    std::cout << "Sum of " << *ptrNum1 << " and " << *ptrNum2 <<
    " is: " << *ptrSum << std::endl;

    return 0;
}
```

Q2. Write a function to find out the first and last digit of a number without returning anything.

```
#include <iostream>

// Function to find and print the first and last digit of a
// number using pointers
void findFirstAndLastDigit(int number, int* firstDigit, int*
lastDigit) {
    *lastDigit = number % 10;

    // Find the first digit
    while (number ≥ 10) {
        number /= 10;
    }
    *firstDigit = number;
}

int main() {
    int num, firstDigit, lastDigit;

    // Input a number
    std::cout << "Enter a number: ";
    std::cin >> num;

    // Call the function to find and print the first and last
    // digits using pointers
    findFirstAndLastDigit(num, &firstDigit, &lastDigit);

    // Print the first and last digits
    std::cout << "First Digit: " << firstDigit << std::endl;
    std::cout << "Last Digit: " << lastDigit << std::endl;

    return 0;
}
```



**THANK
YOU !**

